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Digital Transformation as a Leadership Discipline: Analysis, Evidence, and Resources

ITM 530 Final Paper

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Introduction

Digital transformation is the leadership topic I have chosen for this final paper, and the choice is not arbitrary. For over a decade I have been working at the intersection of data strategy, technology leadership, and organisational change in the Nigerian energy sector. The questions this topic addresses, what digital transformation actually requires of organisations, why it consistently underperforms expectations, and what effective leadership of a transformation looks like in practice, are questions I have been living with rather than studying from a distance. That proximity sharpens the analysis but also requires intellectual honesty about what the research actually shows versus what practitioners wish it showed.

This paper is organised in two parts. The first analyses two foundational papers on digital transformation in depth, examining their arguments, their evidence, their limitations, and their implications for technology leaders. The second identifies three additional resources that are valuable precisely because they approach the topic differently from the primary sources, offering perspectives that complement and complicate the main argument in useful ways. The goal throughout is not to produce a literature review but to engage critically with research that has real implications for how technology leaders make decisions.

Part 1: Critical Analysis of Two Primary Papers

Paper 1: Verhoef et al. (2021) — Digital Transformation: A Multidisciplinary Reflection and Research Agenda

Verhoef, Broekhuizen, Bart, Bhattacharya, Qi Dong, Fabian, and Haenlein published this paper in the Journal of Business Research in 2021. It is one of the most cited and most comprehensive academic treatments of digital transformation available, and it is worth engaging with seriously precisely because it attempts something ambitious: to synthesise how digital transformation is understood across multiple disciplines including marketing, operations management, information systems, and strategy.

What the Paper Gets Right

Definitional precision is the paper's most important contribution, and the one I find most immediately useful in practice. Verhoef et al. (2021) draw a clear and operationally useful distinction between digitisation, the conversion of analogue information to digital form, digitalisation, the use of digital technology to improve existing processes, and digital transformation, the fundamental reconception of how an organisation creates and captures value. This three-stage framework is more analytically useful than most practitioner treatments of the topic, which tend to use transformation as a catch-all term for any technology investment regardless of its strategic scope.

Equally important is the paper's honest assessment of how rarely genuine transformation actually occurs. The authors argue explicitly that most organisations describing themselves as undergoing digital transformation are in the digitalisation stage, improving existing processes rather than reconceiving their fundamental value creation logic. This is an uncomfortable finding for the many organisations that have invested significantly in transformation programmes, and it is more useful than the celebratory narratives that dominate practitioner literature. In my work at Ikeja Electric over the past decade, this distinction has been operationally clarifying. The smart metering and analytics infrastructure we have built represents meaningful digitalisation. Whether it constitutes transformation in the Verhoef et al. sense depends on whether it has changed the fundamental logic of how we create value for customers and stakeholders, which is a harder question to answer honestly.

Multidisciplinary framing is both a strength and a limitation. The paper's synthesis across marketing, operations, information systems, and strategy perspectives demonstrates that digital transformation is genuinely a cross-functional phenomenon that cannot be adequately understood through any single disciplinary lens. This is important for technology leaders who are trained primarily in one domain and need to understand the full scope of what they are managing. The survey of how each discipline frames the transformation challenge is one of the most useful summaries of the field available in a single document.

What the Paper Misses or Gets Wrong

Leadership is the paper's primary limitation. Despite covering digital transformation from multiple disciplinary perspectives, the paper gives relatively little attention to the specific leadership capabilities and behaviours that determine whether transformation initiatives succeed or fail. The research consistently shows that transformation outcomes are shaped more by leadership quality than by technology quality, yet the paper treats leadership primarily as an environmental condition rather than as a subject of analysis in its own right. For a paper that aims to provide a comprehensive research agenda, the relative silence on what effective digital transformation leadership looks like is a significant gap.

Political economy is also underweighted. Digital transformation in established organisations consistently encounters resistance from people and units whose power, status, and resource allocation are threatened by the changes transformation requires. The paper acknowledges cultural resistance as a barrier but does not engage seriously with the mechanisms through which it operates or the specific leadership interventions required to navigate it. An organisation that reads the Verhoef et al. framework as its guide to transformation will understand the stages of the journey and the domains it must address, but will find limited guidance on the organisational dynamics that most consistently derail transformation efforts.

Non-Western and non-corporate contexts represent a third gap in the paper. The empirical basis for the framework draws heavily on studies of large Western enterprises, particularly in retail and financial services. The applicability of the framework to organisations operating in infrastructure-constrained, regulatory-complex, or resource-scarce environments, which describes much of the global economy, is assumed rather than tested. Technology leaders working in contexts like the Nigerian energy sector cannot simply apply the framework without adapting it to the specific conditions that shape what transformation means and what it requires in their context.

Pros and Cons Summary

Primary strengths are definitional rigour, disciplinary breadth, and intellectual honesty about the gap between transformation rhetoric and transformation reality. The three-stage digitisation-digitalisation-transformation framework is genuinely useful for practitioners trying to diagnose where their organisation actually is versus where it claims to be. The research agenda it proposes is substantive and well-grounded.

Primary weaknesses are the limited treatment of leadership dynamics, the underweighting of political and organisational resistance mechanisms, and the narrow empirical base that limits applicability to non-Western and non-corporate contexts. The paper would be strengthened by deeper engagement with how transformation leaders actually build the conditions for change rather than simply identifying what conditions are required. For a technology leader using this paper as a practical resource, those gaps require supplementation from other sources.

Paper 2: Baptista et al. (2020) — Digital Work and Organisational Transformation: Emergent Digital/Human Work Configurations in Modern Organisations

Baptista, Stein, Klein, Watson-Manheim, and Lee published this paper in the Journal of Strategic Information Systems in 2020. Where the Verhoef et al. paper provides a macro-level framework for understanding digital transformation at the organisational level, the Baptista et al. paper examines digital transformation at the level of work itself, asking how the integration of digital systems changes what it means to do work in an organisation and how those changes unfold over time.

What the Paper Gets Right

Baptista et al. (2020) introduce the concept of emergent digital human work configurations as their most original and practically useful contribution. The argument is that digital transformation does not simply automate existing work or replace human labour with digital systems. It creates new forms of work organisation in which humans and digital systems interact in ways that were not designed in advance and cannot be fully anticipated at the point of technology deployment. These configurations emerge through practice, through the accumulated experience of people working with digital systems over time, and they determine whether the technology investment actually produces the organisational capability it was designed to create.

This argument has direct implications for how technology leaders think about implementation. If digital human work configurations emerge through practice rather than being designed in advance, then the focus of implementation management needs to shift from deployment to the ongoing development of human-digital working relationships. The technology going live is not the end of the implementation challenge. It is the beginning of the period during which the organisation learns how to use the technology in ways that actually produce value. I have lived this directly in the data analytics work at Ikeja Electric, where the technical deployment of our data infrastructure preceded by years the development of the analytical working practices that allow the organisation to extract value from that infrastructure.

Methodologically, the paper draws on longitudinal qualitative studies of digital transformation in specific organisational contexts, is more grounded in organisational reality than the survey-based and case study approaches that dominate the digital transformation literature. By following organisations through the transformation process rather than studying them at a single point in time, the paper is able to capture the messiness, the setbacks, and the unexpected adaptations that characterise real transformation journeys. This gives the findings a texture and a credibility that more abstracted frameworks often lack.

What the Paper Misses or Gets Wrong

Focusing on the micro-level dynamics of digital work comes at the cost of strategic perspective. The emergent configurations framework is valuable for understanding what happens after a technology is deployed, but it offers limited guidance on how to select which technologies to deploy, how to sequence transformation investments, or how to manage the macro-level organisational change that transformation requires. A technology leader who uses the Baptista et al. framework needs to supplement it with strategic-level analysis that the paper does not provide.

A further limitation I would name is the paper's optimism about emergence. The argument that digital human work configurations develop through practice implies a degree of adaptive capacity in organisations that is not universally present. Some organisations have the psychological safety, the learning culture, and the management support required for productive adaptation to new working configurations. Many do not. The paper acknowledges this variation but does not fully theorise the conditions that determine whether emergent adaptation leads to capability development or to the entrenchment of workarounds that effectively neutralise the technology investment.

A practical limitation is the paper's accessibility. The writing is dense and the conceptual vocabulary is specialised. A practitioner technology leader attempting to use this paper as a direct resource for managing transformation would find the translation from theory to practice demanding. The paper would benefit from more explicit discussion of the practical implications of its findings for technology leaders, including what specifically changes about how implementation should be managed if the emergent configurations framework is taken seriously.

Pros and Cons Summary

Conceptual originality, longitudinal methodological depth, and the grounding it provides for understanding why technology investments so often underperform despite technical success. The emergent digital human work configurations concept names something real that most implementation frameworks fail to capture. For technology leaders trying to understand why their organisation is not extracting value from deployed technology, this paper offers more useful analytical tools than most alternatives.

Limited strategic scope, optimism about organisational adaptive capacity, the optimism about organisational adaptive capacity, and the accessibility challenges that make practical application difficult. The paper works best as a complement to strategic-level frameworks rather than as a standalone guide. For a technology leader, combining the Baptista et al. micro-level analysis with the Verhoef et al. macro-level framework produces a more complete picture of what transformation requires than either paper provides alone.

Part 2: Three Additional Resources for Understanding Digital Transformation

Resource 1: Bican and Brem (2020) — Digital Business Model, Digital Transformation, Digital Entrepreneurship: Is There a Sustainable Digital?

Published in Sustainability in 2020, the Bican and Brem paper earns its place as a resource for digital transformation leadership because it does something that most digital transformation literature avoids: it disaggregates the concept into three distinct phenomena that require different strategic orientations, different organisational capabilities, and different success metrics. The distinction between digital business models, digital transformation, and digital entrepreneurship is not merely taxonomic. It has direct consequences for how technology leaders design strategies, allocate resources, and measure progress.

What makes this a good resource is how it addresses definitional confusion that is operationally costly. Organisations that conflate digital transformation with digital business model adoption consistently set the wrong success criteria and make the wrong investment choices. An organisation that believes it is transforming when it is actually adopting a digital business model will underinvest in the cultural and operational changes that genuine transformation requires. An organisation that mistakes digital entrepreneurship for transformation will build the wrong capabilities and create the wrong governance structures. The Bican and Brem framework provides the conceptual tools to diagnose which of the three a specific organisation is actually doing, which is the prerequisite for doing it well.

Sustainability in digital strategy is a perspective most transformation literature ignores, and Bican and Brem address it. The question of whether there is a sustainable digital is genuinely important for organisations making long-horizon capability investments. Technology generations turn over, competitive advantages erode, and the specific configurations of digital capability that produce advantage today may not produce advantage in five years. The paper's framing of this question is less fully developed than its definitional contributions, but it points toward a research agenda that is important for technology leaders thinking about how to build digital capabilities that are durable rather than merely current. Bican and Brem (2020) are also useful in how they position digital entrepreneurship as a distinct phenomenon rather than a subset of digital transformation. Organisations that treat entrepreneurial digital activity as transformation tend to create governance and accountability structures that are inappropriate for the speed, ambiguity, and experimental nature of entrepreneurial work. Separating the two conceptually allows organisations to design different management approaches for each, which tends to produce better outcomes in both domains.

Resource 2: Ozalp, Eggers, and Malerba (2023) — Hitting Reset: Industry Evolution, Generational Technology Cycles and the Dynamic Value of Firm Experience

Published in the Strategic Management Journal in 2023, the Ozalp, Eggers, and Malerba paper is a valuable resource for digital transformation leadership because it addresses a problem that most transformation frameworks assume away: the relationship between accumulated organisational experience and the ability to navigate technology generation transitions. The paper's central finding, that firm experience is a competitive asset within a technology generation but a liability when the generation resets, has profound implications for how transformation leaders think about their own organisations and their own leadership capabilities.

Ozalp et al. discuss digital transformation in a new and genuinely uncomfortable way. Most transformation literature treats organisational capability and experience as resources to be built and protected. The Ozalp et al. argument suggests that the same capabilities and experiences that make an organisation effective in the current technology generation are the ones that create the greatest resistance to moving to the next one. This is not a theoretical argument. The empirical evidence from the video game industry, where the authors track firm performance across multiple generational technology transitions, provides unusually rigorous support for a pattern that practitioners in many industries will recognise from observation.

For digital transformation leaders specifically, the paper's most important implication is about the composition of foresight processes. If deep experience in the current technology generation systematically biases perception of the next one, then effective foresight requires including voices that are not embedded in the current way of doing things. This is a harder organisational design challenge than it appears, because the people whose voices matter most for foresight, those who can see the next generation clearly, are often not the people who have the organisational standing to influence resource allocation decisions. The paper names this tension clearly without resolving it, which is honest and useful. Ozalp et al. (2023) also make a contribution that is underappreciated in the digital transformation literature: the empirical rigour of the video game industry analysis. Most digital transformation research relies on case studies and surveys that are difficult to replicate and prone to selection bias toward successful transformations. The longitudinal, large-sample study of how firms performed across multiple generational technology transitions provides a level of empirical grounding that is rare in this field and makes the findings considerably more credible than those derived from smaller, more selective datasets.

Resource 3: Nakash and Bouhnik (2022) — Can Return on Investment in Knowledge Management Initiatives in Organisations Be Measured?

Published in the Aslib Journal of Information Management in 2022, the Nakash and Bouhnik paper earns its place as a digital transformation resource by addressing one of the most persistent and least well-resolved challenges in transformation leadership: how to measure the value of investments whose benefits are primarily organisational capability rather than direct revenue or cost reduction. Digital transformation is, fundamentally, a capability-building exercise. The primary outputs are better decisions, faster organisational responses, improved learning capacity, and enhanced ability to deploy future capabilities. None of these map cleanly onto traditional financial ROI metrics.

What makes this a key resource is its honest engagement with the measurement problem rather than offering a simplified formula that pretends the problem is easier than it is. The authors review the existing approaches to ROI measurement for knowledge and capability investments, identify their limitations, and propose a set of alternative measurement frameworks including balanced scorecard approaches, value realisation models, and qualitative impact assessment. The conclusion is not that measurement is impossible but that it requires different methodologies, different timelines, and different definitions of value than organisations typically apply to operational technology investments.

This matters deeply for digital transformation leadership, and I would argue it is the most practically important resource in this paper for any leader trying to secure transformation investment. Organisations that insist on applying traditional financial ROI criteria to transformation investments before approving them will systematically underfund the category of investment that builds the most durable competitive advantage. The Nakash and Bouhnik paper provides the analytical grounding for making that argument to finance and executive leadership, which is a conversation that technology leaders need to be able to have if they are going to secure the investment that transformation requires. The paper does not provide a simple answer but it provides the vocabulary and the framework for a more sophisticated conversation about value measurement than most organisations are currently having. Nakash and Bouhnik (2022) also identify something important about the political dimension of ROI measurement in transformation contexts. When transformation investments cannot be evaluated through standard financial metrics, the allocation decision becomes more susceptible to organisational politics and less anchored to evidence. The authors argue that developing better measurement frameworks is not just an evaluation challenge but a governance challenge. Organisations that cannot measure the value of capability investments will consistently allocate resources based on advocacy quality rather than evidence quality, which systematically disadvantages the most strategically important category of transformation investment.

Conclusion

Five sources analysed across this paper, two primary and three supplementary, together provide a more complete picture of digital transformation than any single source could offer. The Verhoef et al. framework provides the definitional precision and the strategic-level map of the transformation journey. The Baptista et al. work provides the micro-level understanding of how technology actually changes work and why that change cannot be managed purely through deployment. The Bican and Brem piece provides the conceptual tools to distinguish transformation from related but distinct phenomena. The Ozalp et al. paper provides the most uncomfortable and most important insight: that the capabilities that make an organisation effective today may be the ones that most impede its ability to navigate tomorrow. And the Nakash and Bouhnik paper provides the measurement framework for justifying and tracking the capability investments that transformation requires.

What these sources share, and what distinguishes them from most popular treatments of digital transformation, is intellectual honesty about difficulty. None of them suggest that transformation is primarily a technology problem that good technology selection and implementation can solve. All of them point toward the organisational, leadership, and cultural dimensions of transformation as the primary determinants of whether the technology investment produces the outcomes it was designed to create. That convergence across different disciplinary perspectives and different levels of analysis is itself a finding worth noting. Digital transformation succeeds or fails at the intersection of technology capability and human organisation. The technology is increasingly the easier part.

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